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Computer System Organization

1.1 INTRODUCTION

Today, everything around us involves technology, primarily computers. Be it withdrawal of money from the ATM, online shopping, e-learning, ticket reservation, payment of telephone and electricity bills, or searching the internet for information, it has become a necessity for everyone to know about computers and the latest technology. Even a student's life revolves around smartphones, whether it is email, web surfing, downloading or uploading information, connecting to social networking sites, clicking photos, so on and so forth. In short, computer has made a great impact on us and has become an integral part of our lives.

A computer's capability to process, store, and retrieve data and information has made it intrinsic to all kinds of environment—home, office or business. Widespread use of the computer has made it imperative for us to have knowledge about it, its advantages and disadvantages, and its internal structure.

So, technically, a computer is defined as an electronic device that takes input from the user in the form of data and instructions, processes this data and produces result (output) as information.

Commit To Memory (CTM): A computer is an electronic device that accepts a set of instructions in the form of a program, executes it and displays the output to the user.

Thus, working of a computer can be expressed as an equation given below—

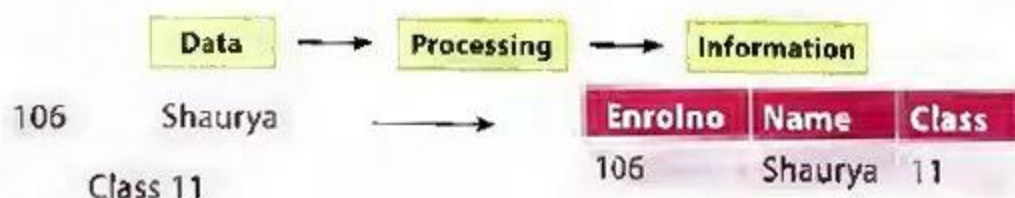


Fig. 1.1: Data, Processing and Information

where data is defined as raw facts or figures such as 106, "Shaurya", "Class 11". This data does not have any meaning when presented as such. However, this data can be organized or processed to transform it into useful information.

Information is defined as a collection of data which is organized in a particular manner to generate meaningful or processed data. For example, "Shaurya is a Class 11 student with Enrolment number 106". This is processed data which gives some meaningful information.

Therefore, a computer takes input in the form of data and generates output in the form of information. This process of converting data into some meaningful information is called **Information Processing Cycle**, which we will discuss now.

1.2 BASIC COMPONENTS OF A COMPUTER SYSTEM

Computer is a combination of hardware and software. Hardware is the physical components of a computer like motherboard, memory device, monitor, keyboard, etc. while software is a set of programs or instructions.

Both hardware and software together make the computer system function. Let us first have a look at the functional components of a computer:

Computer System

Hardware

Software

Fig. 1.2: Components of a Computer System

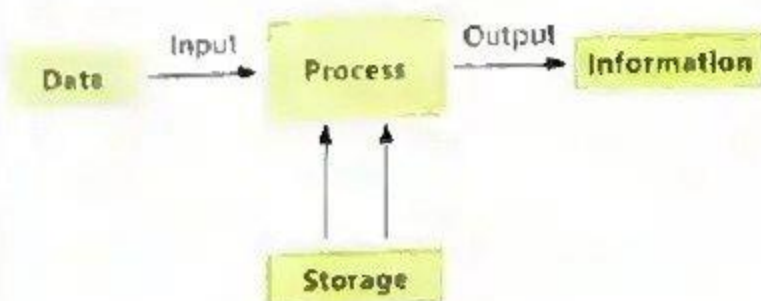


Fig. 1.3: IPO Cycle

Every task given to a computer follows an **Input-> Process-> Output Cycle (IPO cycle)**. It accepts certain input, processes that input and produces the desired output. The input unit takes the input, the **Central Processing Unit (CPU)** processes the data and the output unit produces output. The memory unit holds the data and instructions during the processing.

Fig. 1.4 shows the basic structure of a computer. It is also known as **Von Neumann Architecture**. A computer receives data and instructions through "Input Devices" which get processed by the CPU and the result is shown through "Output Devices". The "Main/Primary Memory" and "Secondary/Auxiliary Memory" are used to store data inside the computer.

The main memory holds the input and intermediate output during the processing. Let us discuss each unit/component in detail.

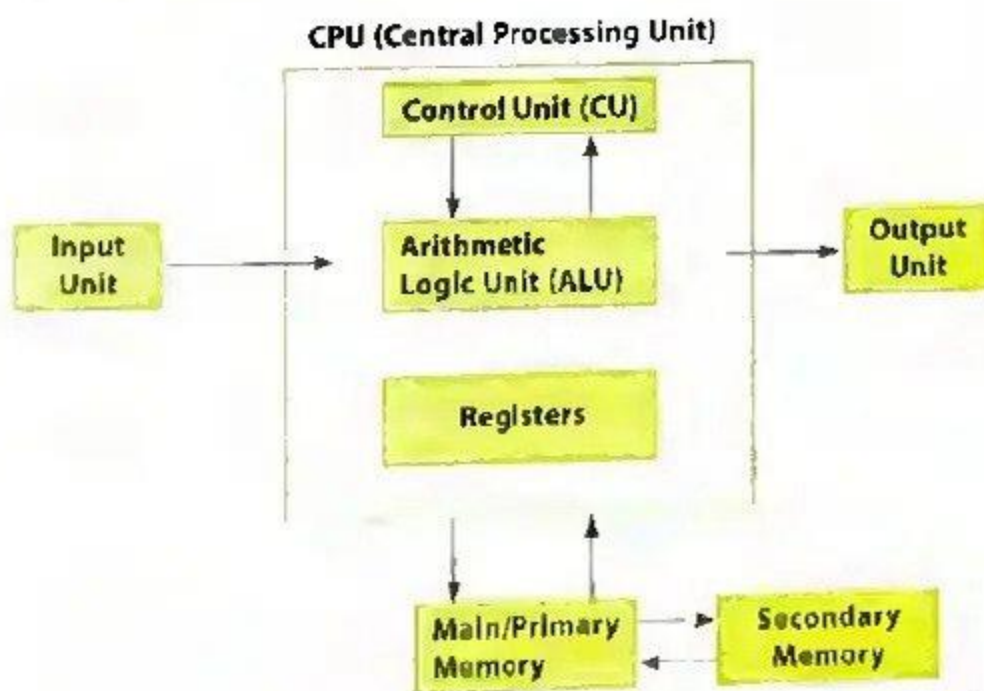


Fig. 1.4: Functional Components of a Computer System (Von Neumann Architecture)

1.2.1 Input Unit

An input unit takes/accepts input and converts it into binary form so that it can be understood by the computer. The computer input constitutes data and instructions.

The input unit comprises input devices attached to the computer. These devices accept input from the user and convert it into a form that can be understood by the computer (i.e., binary code/form). The term "Binary" means two states—ON/OFF or HIGH/LOW—voltage and, in turn, two bits or symbols, 1 for ON and 0 for OFF.

CTM: An input unit fetches the input and converts it into binary form which is directly understood by the computer.

Let us now discuss the most extensively used input devices.

1. **Keyboard:** The most commonly used input device is a keyboard which directly enters the data in the form of letters, digits and commands into the computer.

A computer keyboard possesses additional keys in contrast to a traditional typewriter, such as function keys, alphanumeric keys, direction keys, and other keys (Special and lock keys).



2. **Mouse:** A mouse is a pointing device with a roller at its base, used for moving a pointer on a computer monitor. It converts the movements of the user's hand into a unique set of binary digits representing the position of the mouse at a particular instant. When a user moves a mouse across a flat surface, the pointer moves in the direction of the mouse's movement.

3. **Light Pen:** A light pen is a pointing device that can be used to select anything on the computer screen by simply pointing at it or for drawing figures directly on the screen. It consists of a Photocell mounted on a pen-shaped tube called stylus. It is capable of sensing a position on the screen when its tip touches the screen. Clicking is performed by pressing the pen on the screen. A light pen is mostly used by engineers, architects or designers.



4. **Optical Mark Reader (OMR):** An OMR is capable of recognizing a pre-specified type of mark made with a dark pencil or ink. Special pre-printed forms are designed with the boxes that can be marked with a dark pencil or ink. Such a document is read by an OMR, which transcribes the marks into electrical pulses that are transmitted to the computer. This technology is called Optical Mark Recognition.

It is commonly used for grading specially-designed MCQ answer sheets and in areas where one response is chosen from a small number of alternatives and the volume of data to be processed is very large.

5. **Smart Card Reader:** A smart card is embedded with a microprocessor that can hold a certain amount of personal data in its memory. The special reader machine that is used to read this microprocessor is known as smart card reader. The card is made of plastic, generally PVC. Smart cards are used as ATM cards, ID cards, credit and debit cards. They are used in banking, medical, and in big companies and organizations for strong security authentication.





6. **Bar Code Reader:** A bar code is a collection or a sequence of lines of different heights and widths that are printed on various types of products. The machine that reads these bar codes is called a bar code reader. It consists of a light source, a lens and a light sensor which translates optical impulses into electrical signals. Also, it contains decoder circuitry which analyzes the bar code's image data and sends the bar code's content to the scanner's output port.

7. **QR Code Reader:** QR Code is an abbreviation for Quick Response Code. It is a special kind of barcode that anybody can scan with a Smartphone App that usually directs the user to a website. QR Codes have gained popularity in commercial marketing because they are very easy to handle. The advantage of using QR codes is that you don't have to type an entire web address, the user merely scans the code which is further directed for processing to an associated website, or sent through an SMS message, email address or as an attachment to email message, with a calendar event or even with a location, etc.



8. **Biometric Sensor:** It is an input device that is used to uniquely identify a person on the basis of his physical or behavioural traits. Biometric sensors can scan human characteristics, such as eyes, fingerprints, and DNA with the help of sensors. It is commonly used to mark attendance of employees/students in organizations/institutions. It is also popular as a security device to provide restricted entry to secured areas.

9. **Touch Screen:** It is a type of display screen which allows interaction with computer through a touch-sensitive transparent panel covering the entire screen without any intermediate device. The touch screen uses a technology that enables the user to touch the screen with their fingers to select objects. Touch screens are mostly found in systems such as ATMs and mobile devices where the user needs to choose from a given list. Such systems are also being used at shopping malls, amusement parks and airports.



10. **Microphone:** A microphone is an input device used to provide audio data to a computer. It works with a sound card and is mainly used for sound recording.

11. **Webcam:** A webcam is a camera that is connected to a computer. It captures still pictures as well as videos and, with the help of software, can transmit them in real time. Unlike a digital camera and digital camcorder, a webcam does not have any built-in storage. Instead, it always uses the computer hard drive as its storage.



12. **Magnetic Ink Character Reader (MICR):** MICR detects the numbers printed with magnetically charged ink at the bottom of bank cheques and converts them into digital data. These digital numbers represent the bank account number, bank branch and cheque number.

- 13. Optical Character Reader (OCR):** An OCR allows to recognize scanned images, screenshots, PDF documents, hand-written documents into machine-encoded text. These digitized texts, can be electronically edited, searched or stored compactly and used in machine translation, text-to-speech and other fields of artificial intelligence.



1.2.2 Central Processing Unit (CPU)

The CPU is the control centre/brain of a computer. It guides, directs, controls and governs all the processing that takes place inside the computer. The CPU consists of three components—ALU, CU and Memory unit.

- A. Arithmetic Logic Unit (ALU):** All calculations and comparisons are made in this unit. The ALU performs the arithmetic (+, -, *, /) and logical (AND, OR, NOT, XOR) operations, sent from the memory, performs specific operations (addition, subtraction, etc.) and the result is returned to the memory. The result of the logical operations is either true or false and helps the computer in decision-making.
- B. Control Unit (CU):** The CU controls the flow of data from input devices to memory and from memory to output devices. It controls and guides the interpretation, flow and manipulation of all data and information. It does not actually process the data; instead, it sends control signals to ALU and memory for carrying out the required operations.

Learning Tip: CU controls all the functions like input, output, storage and process. It instructs ALU which operation is to be carried out.

Registers are high-speed temporary storage areas found in the CPU. Registers work as per the instructions given by the control unit (CU), storing instructions and data, immediately required for performing an operation. The CPU places the highest priority jobs/data inside registers for faster execution/processing. Registers can be of different sizes (16 bit, 32 bit, 64 bit and so on) and each register inside the CPU has a specific function like storing data, storing an instruction, storing address of a location in memory, etc.

- C. Memory Unit (MU):** Memory unit (Primary/Main Memory) is used as a storage unit for program as well as data. It is the computer memory that is accessed directly by the CPU. The memory can be both primary and secondary depending upon its location in the computer system. The primary memory, also termed as main memory, is directly accessible to the CPU since all the work is done in the RAM (primary memory) and later on gets stored on the secondary storage (hard disk). Hence, there is an indirect contact between CPU and hard disk. On the contrary, the secondary memory, also known as auxiliary memory or auxiliary storage, includes magnetic storage devices such as hard disk drives, magnetic drums, etc. The secondary storage can be accessed by the CPU through input-output controllers or units. The memory of the computer is like a predefined working space where it temporarily keeps information and data to facilitate its working. When the task is executed or finished, it clears the memory and this memory space is available for the next task to be executed or performed. Main memory, i.e., RAM (Random Access Memory), which is the primary memory of a computer system, is composed of cells. A memory cell is a device that stores a single symbol selected from a set of symbols.

As shown in the Fig. 1.5, each of these cells is further broken down into smaller components—**bits**. A bit stands for binary digit, *i.e.*, either 0 or 1, which is an elementary unit of computer memory. Combinations of these bits together are used to store the data and instructions. Eight bits together form 1 byte.

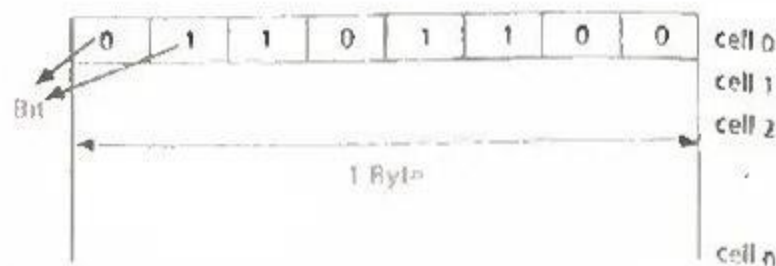


Fig. 1.5: Memory Cells

CTM: 8 bits = 1 byte 4 bits = 1 nibble } Unit of measurement

The memory unit or main memory is divided into:

- Random Access Memory (RAM)
- Read-Only Memory (ROM)

➤ RAM (Random Access Memory)

It is a read/write memory as it is possible to both read from and write to a location within RAM. It is used for primary storage in computers to hold active information of data and instructions. The

RAM is a volatile memory as it does not store data and instructions permanently and loses its contents when the power is switched off or interrupted.

The RAM chips in a computer can be categorized into two types:

- DRAM (Dynamic RAM)
- SRAM (Static RAM)

Static RAM retains its contents as long as the power is connected and is easy to interface but uses six transistors per bit. On the contrary, Dynamic RAM is more complicated to interface and control and needs regular refresh cycles to prevent its contents from being lost. However, DRAM uses only one transistor and a capacitor per bit, allowing it to reach much higher densities and, with more bits on a memory chip, be much cheaper per bit. SRAMs are faster than DRAMs and that is why are used in cache memory while DRAMs are used in main memory. RAM provides faster access than secondary memory with less memory access time.



Fig. 1.6: RAM (Random Access Memory)

CTM: Memory Access Time: The amount of time taken to retrieve data required from memory, from the start of access until the data becomes available, is called memory access time.

➤ ROM (Read-Only Memory)

It is a read-only memory, *i.e.*, the data and instructions are placed in the ROM at the time of its manufacturing and can't be changed thereafter. It is a permanent and non-volatile memory as it does not lose its contents when the power is switched off or interrupted.

ROM is slower than RAM and is used to hold certain essential instructions to check basic hardware components such as booting, procedures to load operating system and frequently needed functions.

The different types of ROM available are:

- PROM (Programmable Read-Only Memory)
- EPROM (Erasable Programmable Read-Only Memory)
- EEPROM (Electrically Erasable Programmable Read-Only Memory)



Fig. 1.7: ROM (Read-Only Memory)

➤ Cache Memory

Cache memory is a small memory that operates much faster than the primary memory or RAM. When the CPU requires certain data present in RAM (primary memory), it first sends the request to the cache [Fig. 1.8(a)], which stores recently-used values. Since cache memory is placed between CPU and main memory, it makes this data available to the CPU very quickly by sending the data to CPU instead of the main memory and thus reduces access time significantly, thereby improving the overall performance of the computer. It is also called CPU memory that a computer microprocessor can access more quickly than it can access regular RAM [Fig. 1.8(b)]. This memory is typically integrated directly with the CPU chip or placed on a separate chip that has a separate bus interconnected with the CPU.



Fig. 1.8(a): Cache Memory

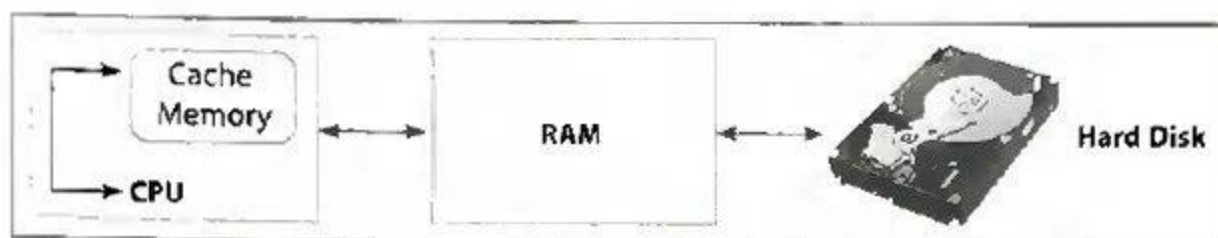


Fig. 1.8(b): Placement of Cache Memory Inside CPU

CTM: Cache Memory, also called CPU memory, is a high-speed memory available inside the CPU in order to speed up access to data and instructions stored in RAM.

• Units of Memory

The elementary unit of memory is a bit. A bit stands for binary digit, i.e., either 0 or 1, which is an elementary unit of computer memory. Combinations of these bits together are used to store the data and instructions. Eight bits together form 1 byte and a group of four bits is called a Nibble.

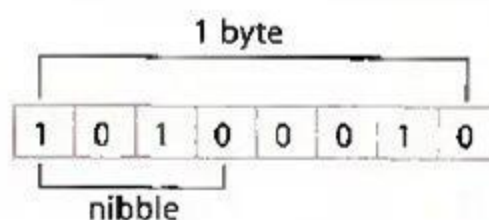


Table 1.1 shows the computer memory units used till date.

Table 1.1: Computer Memory Units

S. No.	Unit	Description
1.	Binary Digit	1 Bit
2.	Byte	1 Byte = 8 Bits
3.	Kilo Byte (KB)	1 KB = 1024 Bytes
4.	Mega Byte (MB)	1 MB = 1024 KB
5.	Giga Byte (GB)	1 GB = 1024 MB
6.	Tera Byte (TB)	1 TB = 1024 GB
7.	Peta Byte (PB)	1 PB = 1024 TB
8.	Exa Byte (EB)	1 EB = 1024 PB
9.	Zetta Byte (ZB)	1 ZB = 1024 EB
10.	Yotta Byte (YB)	1 YB = 1024 ZB
11.	Bronto Byte (Brontobyte)	1 Bronto Byte = 1024 YB
12.	Geop Byte (Geopbyte)	1 Geop Byte = 1024 Brontobytes

1.2.3 Output Unit

Output unit is formed by the output devices attached to the computer. Output devices produce the output generated by the CPU in human readable form. These devices can also be used to store the result for further use.

The commonly used output devices are explained as under:

- **Visual Display Unit (VDU)/Monitor:** The monitor, popularly known as screen or digital projector, is the most common device for displaying the output of the computer-processed information. It displays information in the same way as it is seen on a television screen. The monitor is also called Visual Display Terminal (VDT) or Visual Display Unit (VDU). Its display may be CRT (Cathode Ray Tube), LCD (Liquid Crystal Display), Light-Emitting Diode (LED) and Organic Light-Emitting Diode (OLED).



- **LCD Screen (Television):** A Liquid Crystal Display (LCD) is smaller and lighter in weight as compared to a CRT monitor and, hence, ideal for use in laptops, palmtops and other portable devices.



- **LED Screen:** A Light-Emitting Diode is a light weight flat-panel display and uses light-emitting diodes to create pixels on the screen. LED monitors use less power as compared to CRTs and LCDs, and are considered environment-friendly.



- **OLED:** This technology is more advanced than LED. OLED uses an organic substance which glows when an electric current is passed. This revolutionary material reduces the thickness and weight of the TV, creating an ultra-thin, sleek form, superior colour, reproduction, and individual pixels that switch off for true blacks.



- **Printer:** A printer is an output device which is used to generate hard copies (printout) of the output generated by the computer system. The printer can generate both text and images on paper.

Printers are classified as **Impact** (there is a mechanical contact between printer head and paper) and **Non-impact Printers** (no mechanical contact between printer head and paper). The various types of printers are as follows:

- Dot Matrix Printer:** A dot matrix printer (also known as Serial printer) prints one character at a time. It uses dots to create an image. This printer prints characters by striking an ink-soaked ribbon against the paper which produces sound and, hence, is termed as Impact Printer. These printers have low operating costs and can be used to generate carbon copies also. They can print one line of text at a time and can only print low-resolution graphics.



(ii) **Inkjet/Deskjet/Bubble Jet Printers:** An inkjet printer is the most common type of low-cost printer. It uses the technique of spreading quick dry ink on paper. The ink is stored in the form of cartridges of different colours (red, green, black and yellow). These printers generate high quality prints and are ideal for small offices and homes.



(iii) **Laser Printers:** These printers use laser technology to produce printed documents. These are very fast printers and are used for high quality prints.



➤ **Speakers:** A speaker is a type of output device that generates sound as an output. For a speaker to produce sound, a special device called sound card is required to be installed in the computer system.

➤ **Plotters:** Plotters are the output devices that are used for producing good quality images and drawings. Unlike printers, they support printing of large-sized papers. They are mainly used in computer-aided designing (CAD).



CTM: A pixel is the smallest element of an image on a computer display. Pixel is short for picture element.

1.3 SECONDARY MEMORY

Secondary storage devices are used to store a large amount of data permanently, which is not possible by using the primary or main memory. The amount of data a disk can hold is defined as Disk Capacity, which is measured in terms of bytes, kilobytes (KB), megabytes (MB), and so on. Some of the common secondary storage devices are as follows:

(a) **Hard Disk:** A hard disk is a non-volatile, high-capacity storage device ranging from 1GB to several Terabytes. It consists of solid rounded disks, packed on one another, made up of a magnetic material and sealed inside a case. A hard disk does not get lost or damaged by mishandling as it is generally fixed inside the computer.

Data is stored on the platters in tracks, sectors and cylinders to keep it organized and easier to find.

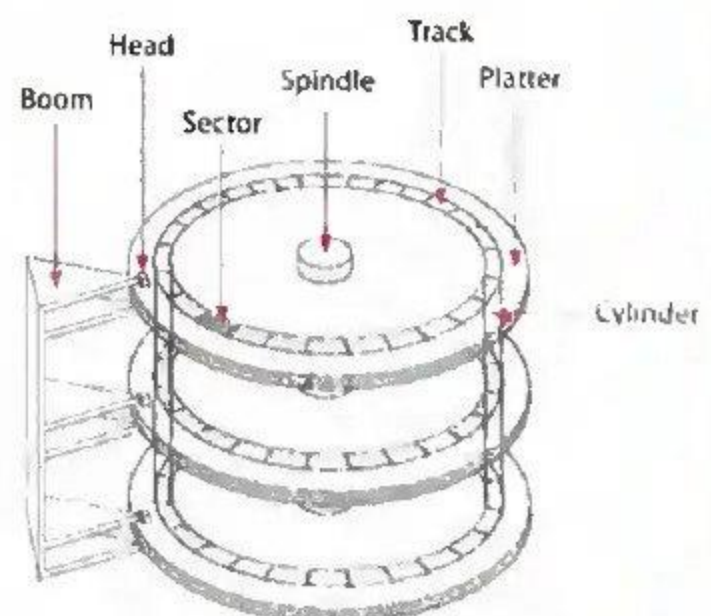
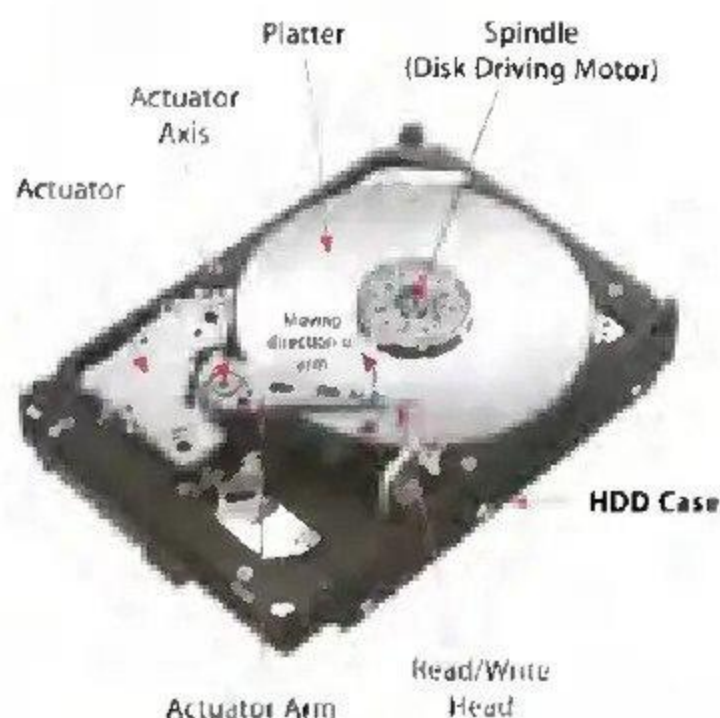


Fig. 1.9: Hard Disk and its Major Components

- **Track:** Each platter is divided into concentric rings called "Tracks". There are thousands of tracks on each platter. A track is divided into segments of **sectors** which is the basic unit of storage.
- **Sector:** Each track is divided into sectors that actually store the data. A sector, as a rule, holds 512 bytes of data.
- **Cylinder:** A cylinder consists of a set of tracks described by all the heads (on separate platters) at a single seek position. Each cylinder is equidistant from the centre of the disk.

- (b) **Blu-Ray Disc:** A Blu-ray Disc (BD) is a high-capacity optical disk medium developed for recording, rewriting and playing back high definition video. It uses blue rays that permit large amount of data to be stored at a greater density. A single layer disc can hold 25 GB while a dual layer can hold 50 GB of data. It was designed to supersede the DVD. Blu-ray discs support higher resolutions and more advanced video and audio formats as compared to DVDs.



- (c) **Compact Disk (CD):** It is a thin optical disk which is commonly used to store audio and video data. The capacity of a standard 120 mm CD is 700 MB. Originally, CD-ROM drives had transfer rate of only 150 KB/s which was very low. The latest CD-ROM drives can transfer up to 72x, i.e., 10800 KB/sec.

- (d) **Magnetic Tapes:** In this, magnetic coatings are stored as data on a thin tape. Data read/write speed is slower because of sequential access. Magnetic tapes are convenient, secure and affordable. Companies across the world still rely on this technology to keep their records safe.



- (e) **DVD:** Digital Versatile Disc or Digital Video Disc is an optical disc storage device. It can be recorded on one or both sides. Its capacity may range from 4.7 GB to 8.5 GB.

- (f) **USB Pen Drive:** Generally called pen drive, this is a small, portable memory which can be plugged into a computer with USB Port. A pen drive's capacity is less than that of the hard disk but much higher than a floppy or CD. It is more reliable also. Pen drives have a storage capacity of 2 GB, 4 GB, 8 GB, 16 GB, 32 GB and 64 GB.



- (g) **Memory Cards:** These are data storage devices, also called Flash memory cards. They are mainly used with digital cameras, computers, mobile phones, music players, video game consoles, etc. They offer high recording ability with power-free storage. Memory cards come with a storage capacity of 8 GB, 16 GB, 32 GB, 64 GB and 128 GB.

1.4 COMMUNICATION BUS—DATA TRANSFER BETWEEN MEMORY AND CPU

In computer architecture, a bus is a collection of wires that transfers data between computer components, i.e., carries binary information to or from input/output devices and memory. It usually transmits binary numbers, one bit per wire.

The bus system in computer is made up of three types of buses:

1. Address Bus
2. Data Bus
3. Control Bus

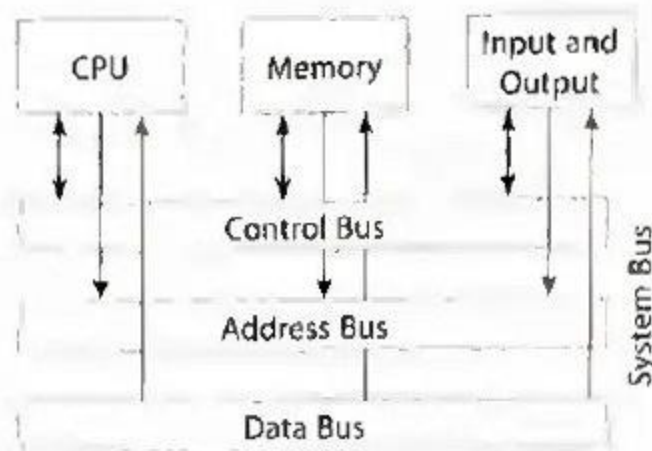


Fig. 1.10: Communication Bus System

1. **Address Bus:** Address bus system is used to specify address of a data/memory location. CPU is connected to the main memory by a set of parallel wires which is the Address bus, which carries address to Memory Address Register (MAR). The width of a bus determines the number of memory locations that can be addressed. Address bus consists of 16 wires, thus, it consumes 16 bits, i.e., its width is 16 bits. For example, a 64-bit address bus can transfer 2^{64} memory locations/addresses.
2. **Data Bus:** Data bus carries data in binary form. It is an 8-bit bus to transfer data from one place/component to another in a computer system. The size of data bus from memory to CPU equals the number of bits in an instruction, called CPU word length. Also, the number of parallel wires is called bus width as it depends on the number of bits.
3. **Control Bus:** Control bus carries instructions to carry out operations such as Read from/Write to memory and its associated Input-Output (I/O) operations.

Apart from the above described important bus systems, a separate type of bus called I/O (Input-Output) bus connects the input, output and other external devices to the system.

1.5 SOFTWARE CONCEPTS

A computer system has two parts—**Hardware** and **Software**. Hardware refers to the physical or tangible (which can be touched) components while software is the code; in other words, a set of instructions necessary to operate/run the computer hardware.

We can also say that software is a set of programs which is designed to perform a well-defined function. A program is a sequence of instructions written to solve a particular problem and to make the hardware run. All the programs used in a computer to perform specific tasks are called software.

Types of Software

Software can be broadly categorized into four types depending upon their functions and user requirement:

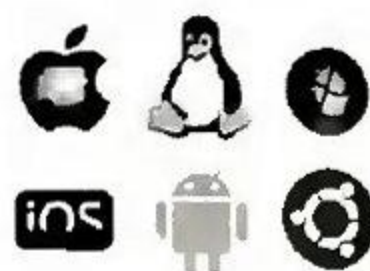
1. System Software
2. Application Software
3. Utility Software
4. Programming Tools



Fig. 1.11: Classification of Computer Software

1.5.1 System Software

These programs provide the environment for running application programs and are designed to control the operation of a computer system. System software comprises programs which interact with the hardware at the core or basic level. It directs the computer what to do, when to do and how to do. Thus, system software serves as the interface between hardware and the user.



The different functions of system software are:

1. Reading data and receiving information
2. Translating data and instructions
3. Controlling all the peripheral devices
4. Processing and generating output

System software can be further categorized into

1. Operating System
2. Language Translators

CTM: The software that controls internal computer operations is called **System Software**.

Concept of Operating System

An operating system is a program that acts as an interface between the user and hardware of the computer as shown in Fig. 1.12. The OS helps to manage resources of the computer and optimize its performance. An OS is the first program to be executed on a computer after the BIOS. OS performs all basic tasks such as identifying basic input/output devices, accepting input from the input devices, sending results to the output devices, keeping track of files and directories on the disk, and controlling other peripheral devices such as disk drives and printers.

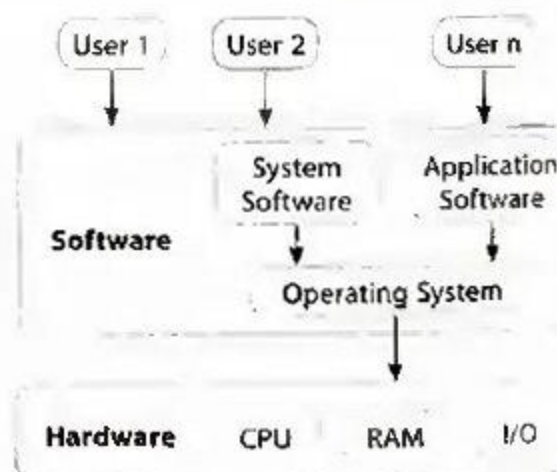


Fig. 1.12: Operating System

CTM: An operating system is an integrated set of specialized programs that is used to manage the overall operations of a computer. It acts like an interface between the user, computer hardware and software.

Every computer must have an operating system to run other programs. DOS (Disk Operating System), UNIX, LINUX and Windows are some of the commonly-used operating systems.

➤ Need for an Operating System

Operating system is an essential component of the system software inside a computer system. An operating system is needed for the following reasons:

1. **User Interface:** Operating system provides instructions to prepare user interface, i.e., way to interact with the user either through GUI (Graphical User Interface) or command prompts.
2. **Program Execution:** It loads necessary programs into the computer memory which are required for its proper functioning.
3. **Resource Allocation:** Operating system controls and allocates the system resources like CPU time, memory (RAM), etc.
4. **Manipulation of File System:** It manages the method/format in which information is stored on and retrieved from the hard disk.
5. **I/O Operations:** Operating system handles all the input-output (I/O) operations.
6. **Error Detection:** It performs the crucial function of error detection and handling.
7. Operating system controls the various system hardware and software resources and allocates them to the users or programs as per their requirement.

Thus, an operating system performs several functions such as Process Management, Memory Management, File Management and Device Management, which we will now discuss in detail.

➤ Functions of OS

The main task an operating system carries out is the allocation of resources and services, such as allocation of memory, devices, processors and information. The major functions performed by an Operating System are as follows:

1. Resource Management

Operating system is responsible for allocating resources to specific programs to complete their tasks.

When a computer has multiple users, the need for managing and controlling the resources (like memory, file, devices, etc.) is even greater. Operating system keeps track of who is using which resource, grants resource request and handles the same request from different users and programs.

Thus, operating system as a resource manager manages and protects multiple computer resources: CPU, Processes, Internal/External memory, Tasks, Applications, Users, Communication channels, etc.

It handles and allocates resources to multiple users or multiple programs running at the same time and space (e.g., processor time, memory, I/O devices).

It also decides between conflicting requests for efficient and fair resource use (For example, maximize throughput, minimize response time). Resource management constitutes multiplexing (sharing) resources. It is carried out in two different manners:

- Time multiplexing
- Space multiplexing

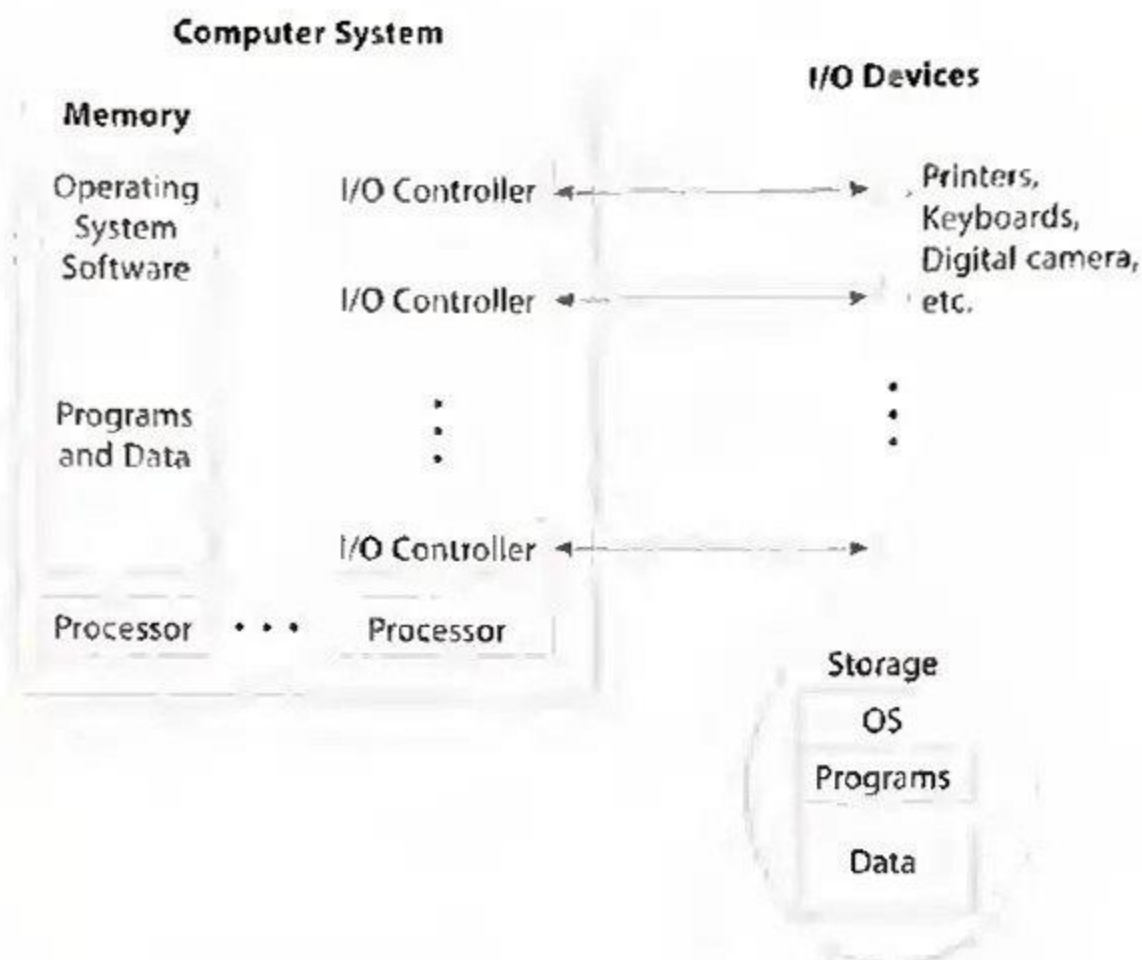


Fig. 1.13: OS as a Resource Manager

In time multiplexing, CPU time gets shared among different programs or users use it in turns. Consider a situation where different programs give command to print at the same time. Time resource manager manages the task and determines who will go next and for how long. Programs or users take turns, i.e., only one at a time. Thus, the multiple print commands lined up on a single printer shall be carried out one by one.

In space multiplexing, instead of completing one by one, each one gets some part of the resource. Main memory is divided amongst several running programs. OS assumes enough memory to hold multiple programs as it is more efficient to hold several programs at once rather than allocating all the memory to a single user.

2. Process Management

While a computer system is operational, different tasks are running simultaneously. A program is intended to carry out various tasks. A task in execution is known as process. We can activate a system monitor program that provides information about the processes being executed on a computer. In some systems it can be activated using Ctrl+Alt+Delete. It is the responsibility of the operating system to manage these processes and get multiple tasks completed in minimum time. As CPU is the main resource of computer system, its allocation among processes is the most important service of the operating system. Hence, process management concerns the management of multiple processes, allocation of required resources, and exchange of information among processes.

3. Memory Management

Primary or main memory of a computer system is usually limited. The main task of memory management is to give (allocate) and take (free) memory from running processes. Since there are multiple processes running at a time, there arises a need to dynamically (on-the-go) allocate and free memory to the processes. Operating

system should do it without affecting other processes that are already residing in the memory and once the process is finished, it is again the responsibility of the operating system to take the memory space back for reutilization. Hence, memory management concerns the management of main memory so that maximum memory is occupied or utilized by a large number of processes while keeping track of each and every location within the memory as free or occupied.

4 File Management

Data and programs are stored as files in the secondary storage of a computer system. File management involves creation, updating, deletion and protection of these files in the secondary memory. Protection is a crucial function of an operating system as multiple users can access and use a computer system. There must be a mechanism in place that will stop users from accessing files that belong to some other user and have not been shared with them. File management system manages secondary memory while memory management system handles the main memory of a computer system.

5. Device Management

A computer system has many I/O devices and hardware connected to it. Operating system manages these heterogeneous devices that are interdependent. The operating system interacts with the device driver and the related software for a particular device. The operating system must also provide the options for configuring a particular device so that it may be used by an end-user or some other device. Just like files, devices also need security measures and their access to different devices must be restricted by the operating system to the authorized users, software and other hardware only.

OS User Interface

OS provides an interface to allow communication between computer hardware and the user. There are different types of user interfaces, each of which provides a different functionality. Some commonly-used interfaces are shown in Fig. 1.14.

(a) Command-based Interface

Command-based interface provides the user to type commands to instruct the OS to perform the desired actions. Using command-based interface, a user can perform several actions such as creating, opening, editing or deleting a file. The user has to remember the names of all such programs or specific commands which the operating system supports. The primary input device used by the user for command-based interface is the keyboard. Command-based interface is often less interactive and usually allows a user to run a single program at a time.

Examples of operating systems with command-based interface include MS-DOS and Unix.

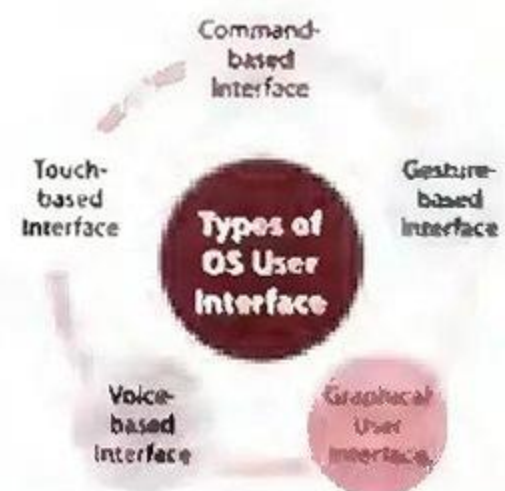


Fig. 1.14: Types of User Interface of OS

(b) Graphical User Interface

Graphical user Interface is a type of user interface that allows a user to interact with computer through images rather than text commands in the form of icons, menus and other visual options. Icons usually represent files and programs stored on the computer and windows represent running programs that the user has launched through the operating system. The input devices used to interact with the GUI commonly include the mouse and the keyboard. Examples of operating systems with GUI interfaces include Microsoft Windows, Ubuntu, Fedora and Macintosh, among others.

(c) Touch-based Interface

Today smartphones, tablets and PCs allow users to interact with the system simply using the touch input. Using the touchscreen, a user provides inputs to the operating system which are interpreted by the OS as commands like opening an app, closing an app, dialling a number, scrolling across apps, etc. Examples of popular operating systems with touch-based interfaces are Android and iOS. Windows 8.1 and 10 also support touch-based interfaces on touchscreen devices.

(d) Voice-based Interface

Modern computers have been designed to address the needs of all types of users including people with special needs and people who want to interact with computers or smartphones while doing some other task. For users who cannot use the input devices like the mouse, keyboard, and touchscreens, modern operating systems provide other means of human-computer interaction. Users today can use voice-based commands to make a computer work in the desired way. Some operating systems which provide voice-based control to users include iOS (Siri), Android (Google Assistant or "OK Google"), Microsoft Windows 10 (Cortana) and so on.

(e) Gesture-based Interface

Some smartphones based on Android and iOS as well as laptops let users interact with the devices using gestures like waving, tilting, eye motion and shaking. This technology is evolving faster and it has promising potential for application in gaming, medicine and other areas.

Language Translator/Compilation

We know that computer understands instructions in machine code, i.e., in the form of 0s and 1s. It is difficult for us to write a computer program directly in machine code. The programs are written mostly in high-level languages, i.e., BASIC, C++, Python, etc. A program written in any high-level programming language (or written in assembly language) is called the Source Program or Source Code.

The source code cannot be executed directly by the computer. It must be converted into machine language to be executed. The program translated into machine code is known as Object Program or Object Code.

The special translator system software that is used to translate the program written in high-level language (or assembly language) into machine code is called **language processor** or **translator program**. The language processors can be of any of the following three types—Assembler, Compiler and Interpreter.

(a) Assembler

Assembler is used to translate the program written in assembly language into machine code. The input of assembler is a source program that contains assembly language instructions. The output generated by the assembler is the object code or machine code understandable by the computer.

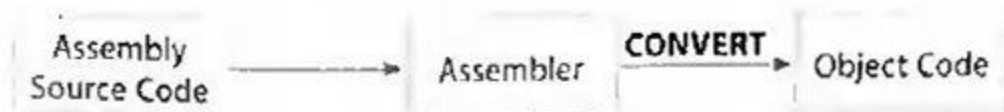
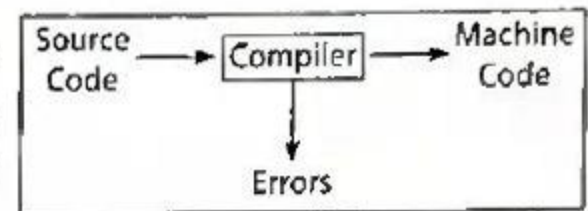


Fig. 1.15: Working of an Assembler

(b) Compiler

The language processor that translates the complete source program as a whole in one go into machine code is called compiler. Some of the examples are C and C++ compilers.



The program translated into machine code is called the object program. The source code is translated into object code successfully if it is free from errors. If there are any errors in the source code, the compiler specifies the errors at the end of the compilation with line numbers. The errors must be removed before the compiler can successfully recompile the source code again.

(c) Interpreter

The language processor that translates a single statement of source program into machine code and executes it immediately before moving on to the next line is called an Interpreter. If there is an error in the statement, the interpreter terminates its translating process at that statement and displays an error message.

Only after removal of the error does the interpreter move on to the next line for execution. Some of the languages that use interpreter are Python, PHP, MATLAB, etc.

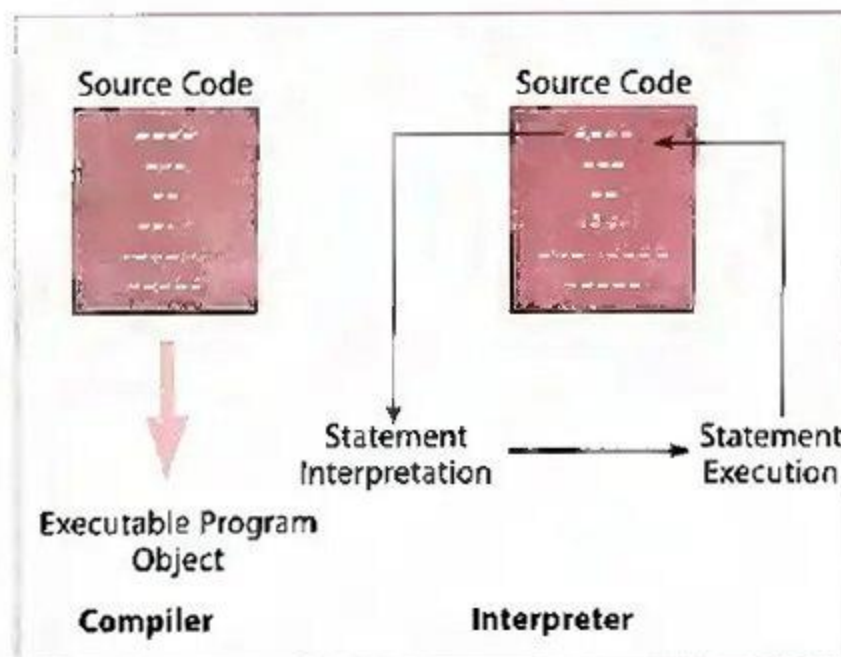


Fig. 1.16: Compiler vs Interpreter

➤ Difference between Compiler and Interpreter

1. A compiler debugs the whole program in one go whereas an interpreter debugs it line by line.
2. Errors traced by a compiler are displayed at the end of the program along with their respective line numbers. On the other hand, interpreter displays the errors line-wise. It does not move on to the next line or execute further until the error in the current line is removed.
3. A compiler program occupies more memory space as it generates an executable file (byte code) in comparison to an interpreter. The entire object code generated by the compiler has to reside in memory and need not be compiled every time. But in the case of an interpreter, it results in more memory wastage since the program is interpreted line-by-line every time the program is executed.
4. A compiler takes less execution time in comparison to an interpreter.

1.5.2 Application Software

Application (app, for short) software is bought by the user to perform specific applications or tasks, *for example*, making a document or making a presentation or handling inventory or managing the employee database.

Application software are classified into two types—General Purpose Application Software and Customized Application Software.

➤ General Purpose Application Software

Some of the application software are made for the common users for day-to-day applications and uses. These are also referred to as Office Tools. Users may use them in the manner they want. Application software developed for generic applications to cater to a bigger audience in general are called general purpose software. Such ready-made application software can be used by end-users as per their requirements. *For example*, spreadsheet tool LibreOffice Calc can be used by any computer user to do calculation or to create an account sheet. Adobe Photoshop, GIMP, Mozilla web browser, iTunes, etc., fall in the category of general purpose software.

➤ Specific Purpose Application/Customized Software

Customized Software is one which is tailor-made as per the user's requirement. Such type of software is customer-specific. It is similar to buying a piece of cloth of a specific colour and fabric and getting it stitched as desired. It is made keeping in mind the individual needs of the user and so is also referred to as Domain Specific Tool. Such software cannot be installed and used by any other user/customer since the requirements may differ. *for example*, Banking System, Payroll Management System, Financial Accounting, Inventory Management Software, Billing System, etc.

POINT TO REMEMBER

A computer system can work without application software, but it cannot work without system software. *For example*, we can use a computer even if no word processing software is installed, but if no operating system is installed, we cannot work on the computer. In other words, the use of computer is possible in the absence of application software.

1.5.3 Utility Software

Utility software are those application programs that assist the computers by performing housekeeping functions like scanning, data backup, removing viruses, etc.

Utility software provides certain tasks that help in proper maintenance of the computer. The job of utility programs is to help the computer system run smoothly. Nowadays, many utility software are part of the operating system itself. Even if there are no utility software on your computer, the computer will work, but with the right kind of utility software loaded, the computer becomes more reliable and even its processing speed increases. Some of the commonly used utility software are antivirus, disk defragmenter, backup, compression, etc.

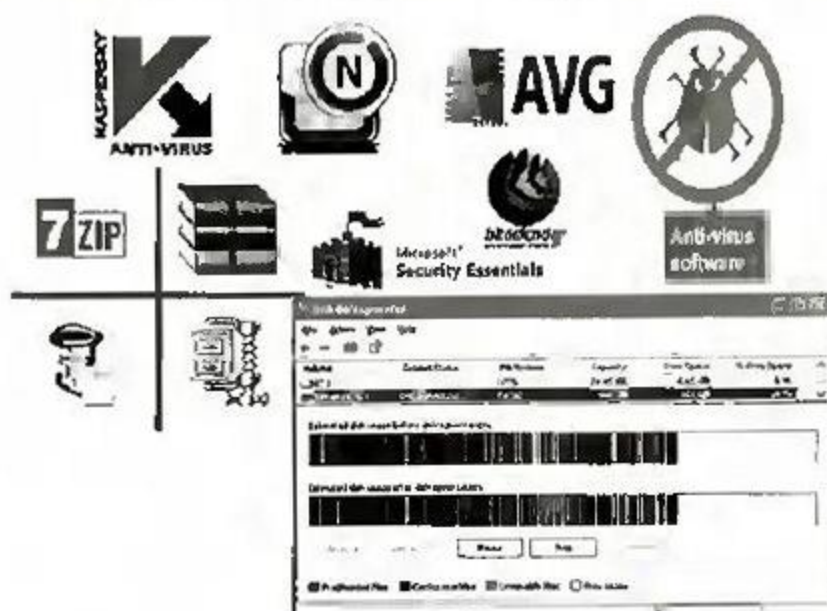


Fig. 1.17: Utility Software

➤ Antivirus Software

An antivirus is a utility software which detects and removes computer viruses or infected areas from the computer hardware for smooth functioning. If the software is not able to remove the virus, it is neutralized. The antivirus keeps a watch on the functioning of the computer system. If a virus is found, it may alert the user, flag the infected program or kill the virus.

➤ Disk Defragmenter

This utility helps to rearrange files and unused space on the hard disk. The memory is used in small chunks randomly. Sometimes when a memory chunk of appropriate size is not available, the operating system breaks or fragments the files, resulting in slower access to files. A disk defragmenter scans the hard disk for fragmented files and brings all the fragments together.

➤ Backup Utility

This utility helps in taking backup of the data, i.e., duplicating the disk information. It creates a copy of the complete or partial data stored in a disk or DVD or CD on any other external disk attached to it. In case the hard disk crashes or some other system failure occurs, the files can be restored using backup software.

➤ Compression Utility

This utility helps in compression of large files. Using compression, files are stored in a special format that takes less space.

Moreover, these compressed files can be restored to their original form when required. Thus, compression is useful because it helps reduce resource usage and reducing the file size makes file transmission on the network easier.

➤ Disk Cleaner

This utility scans for files that have not been accessed/used since long. Such files might be occupying a huge amount of memory space. In that case, the Disk Cleaner utility prompts the user to delete such files so as to create more space on the disk. If the files are important, the user might take a backup before deleting them.

➤ Device Drivers

A device driver is a **program** that controls a particular type of device (usually a peripheral device) that is externally attached to the computer. There are device drivers for printers, scanners, displays, web cameras, modems, DVD readers and so on. Nowadays, most of the device drivers are inbuilt and we don't have to specially install them.

1.5.4 Programming Tools

Working with a computer system is always in the form of IPO—Input, Process, Output. The set of instructions given to the computer to perform any particular task is termed as Program and acts as an input to obtain the desired result. These instructions are written with the help of programming languages, broadly classified as programming tools. There is a remarkable difference between the languages which the computer understands and the languages in which the user works. While humans are able to write programs in high-level language, computers understand machine language.



There is a continuous need for conversion from high-level to machine-level language for which translators are needed. Also, to write instruction, code editors (e.g., IDLE in Python) are needed. We will briefly describe here the programming languages, language translators and program development tools.

(A) Classification of Programming Languages

A user cannot write instructions in the form of 1s and 0s. Thus, bifurcation on the basis of computer programming languages has been made to simplify the process of coding. Two major categories of computer programming languages are low-level languages and high-level languages.

Low-level languages are machine dependent languages and include machine language and assembly language. Machine language uses 1s and 0s to write instructions which are directly understood and executed by the computer. But writing a code in machine language is difficult as one has to remember all operation codes and machine addresses. Also, finding errors in the code written in machine language is difficult. To simplify the writing of code, assembly language was developed that allowed usage of English-like words and symbols instead of 1s and 0s. But one major drawback of writing a code in this language is that the code is computer-specific, i.e., the code written for one type of CPU cannot be used for another type of CPU.

High-level languages are machine independent and simpler to write a code. Instructions are in English-like sentences and each high-level language follows a set of rules, similar to natural languages.

However, these languages are not directly understood by the computer. Hence, translators are needed to translate high-level language codes into machine language. Examples of high level language include C, C++, Java, Python, etc.

(B) Language Translators

As computer can understand only machine language, a translator is needed to convert a program written in assembly or high-level language to machine language. The program code written in assembly or high-level language is called source code. The source code is converted by a translator into machine-understandable form called object (machine) code (Language Translators have been explained in detail under system software section 1.5.1.)

(C) Program Development Tools

Whenever we decide to write a program, we need a text editor. An editor is a software that allows us to create a text file where we type instructions and store the file as the source code. Then an appropriate translator is used to get the object code for execution. In order to simplify program development, there are software called Integrated Development Environment (IDE) consisting of text editor, building tools and debugger. A program can be typed, compiled and debugged from the IDE directly. Besides Python IDLE, NetBeans, Eclipse, Atom and Lazarus are few other examples of IDEs. Debugger, as the name implies, is the software to detect and correct errors in the source code.



MEMORY BYTES

- **Data:** Data is defined as raw facts, such as figures, symbols and numbers that can be used for analysis.
- **Information:** Information is defined as a collection of data. Processed data is information.
- **Hardware devices:** Input devices, system unit, output devices and memory devices are collectively known as hardware devices.
- **Bus:** A bus can be defined as a set of wires/cables to carry binary information to or from input/output devices and memory.
- A microprocessor is a small-sized electronic component inside a computer that performs basic arithmetic and logical operations on data.
- Microcontroller is a small computing device which has a CPU, a fixed amount of RAM, ROM and other peripherals embedded on a single chip.
- The CPU interacts closely with primary storage or main memory, referring to it for both instructions and data.
- There are three basic components in the CPU register section—arithmetic logic unit, memory and control unit.
- Central processing unit is the brain of any computer system.
- Control unit controls all the hardware operations, i.e., input and output units, storage and processor, etc.
- The arithmetic logic units in computers are capable of performing addition, subtraction, division and multiplication as well as some logical operations such as AND, OR, NOT, XOR.
- The memory of a computer can hold program instructions, data values and the intermediate results of calculations.
- Memory Unit is an essential component for storing the programs and data executed by the CPU.
- The main memory of a computer is also known as RAM, which stands for Random Access Memory.
- External memory, which is sometimes called backing store or secondary memory, allows permanent storage of large quantities of data.
- Software are a set of instructions written to achieve the desired tasks and are mainly categorized as system software, programming tools and application software.
- Hardware of a computer cannot function on its own. It needs software to be operational or functional.

- Operating system is an interface between the user and the computer and supervises the working of computer system, i.e., it monitors and controls the hardware and software of the computer system.
- The numbers '0' and '1' are called binary digits.
- A buffer is a data area shared by hardware devices or program processes that operate at different speeds or with different sets of priorities.
- Memory card: A memory card can store various types of data, such as text, pictures, graphics, audio and video.
- Operating system is an interface between the user and the computer and supervises the working of computer system, i.e., it monitors and controls the hardware and software of the computer system.

OBJECTIVE TYPE QUESTIONS

1. Fill in the blanks.

- A program which acts as an interface between a user and hardware is called
- A collection of 4 bits is called a
- Binary number system has digits as and
- A collection of 8 bits is called a
- is a pointing device.
- is read-only memory that stores some pre-written instructions.
- is a new universal coding standard adopted by all new platforms.
- The in computers is capable of performing arithmetic and logical operations.
- controls all the hardware operations of a computer system.
- The expanded form of ASCII is
- A is a non-volatile high capacity storage device ranging from 1 GB to several terabytes.
- memory speeds up access to data and instructions stored in RAM.
- is the term for a program being run by the computer.

2. State whether the following statements are True or False.

- Control unit sends control signals to ALU and memory for carrying out the required operations.
- A light pen is a pointing device that can be used to select anything on the computer screen by simply pointing at it or for drawing figures directly on the screen.
- RAM operates much faster than cache memory.
- 1 GB is equal to 1024 KB.
- An operating system acts as an interface between the user, computer hardware and software.
- A computer has the capacity to perform calculations and other logical functions, whereas a calculator only performs arithmetic and geometrical operations.
- A word is the maximum amount of data a CPU can process at once.
- An address bus carries data from one place to another in a computer system.
- Dot matrix printer uses laser technology to produce printed documents.
- The primary memory unit stores the data and instructions permanently.
- ROM is volatile in nature.
- External memory allows permanent storage of large quantities of data.

3. Multiple Choice Questions (MCQs)

- What is the full form of IPO?

(i) Input Process Output	(ii) Input Print Output
(iii) Internet Print Output	(iv) Internet Process Output
- Which is the fastest memory device?

(i) RAM	(ii) Register	(iii) Blu-ray disc	(iv) Hard disk
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- (c) What is the full form of RAM?
(i) Read Access Memory (ii) Random Access Memory
(iii) Raw Access Memory (iv) Right Access Memory
- (d) ROM is a
(i) Volatile memory. (ii) Non-volatile memory.
(iii) Both (i) and (ii) (iv) None of these
- (e) What is information?
(i) Processed data (ii) Collection of programs
(iii) Collection of instructions (iv) None of these
- (f) Which of the following falls under utilities?
(i) Text editor (ii) Backup (iii) Disk defragmenter (iv) All of these
- (g) Which of the following are the sub-units that make the CPU?
(i) Control unit (ii) ALU (iii) Both (i) and (ii) (iv) None of these
- (h) Which is/are a type(s) of OS?
(i) Single-user OS (ii) Multi-user OS (iii) Time-sharing OS (iv) All of these
- (i) What is the full form of ISCI?
(i) International Standard Code for Information Interchange
(ii) Indian Standard Code for Information Interchange
(iii) International Script Code for Information Interchange
(iv) None of these
- (j) 1 TB is equivalent to:
(i) 2^{10} bytes (ii) 2^{10} MB (iii) 2^{10} GB (iv) 2^{10} KB
- (k) Storage of 1 KB means the following number of bytes:
(i) 1000 (ii) 964 (iii) 1024 (iv) 1064
- (l) One megabyte is equivalent to:
(i) 210 bytes (ii) 220 bytes
(iii) 230 bytes (iv) None of these
- (m) Identify the input device(s):
(i) Speaker (ii) Printer (iii) Keyboard (iv) Scanner
- (n) Which of the following is referred to as the brain of computer?
(i) Processor (ii) RAM (iii) Hard Drive (iv) ROM
- (o) is capable of recognizing a pre-specified type of mark made with dark pencil or ink.
(i) OCR (ii) OMR (iii) MICR (iv) Bar Code Reader
- (p) reads a sequence of lines of different heights and widths that are printed on various types of products.
(i) OCR (ii) OMR (iii) MICR (iv) Bar Code Reader
- (q) uniquely identifies a person on the basis of physical or behavioural traits such as fingerprints, DNA, etc.
(i) Touchscreen (ii) Biometric Sensor
(iii) Optical character reader (iv) QR Code

SOLVED QUESTIONS

1. List the names of two components of CPU.

Ans. The names of two components of CPU are as follows:

- Control Unit
- Arithmetic Logic Unit

2. What is an I-P-O cycle?

Ans. A computer takes input in the form of data and generates output in the form of information. This process of converting data into some meaningful information is called Information processing (I-P-O) cycle.

3. List all the functional components of a computer.

Ans. The functional components of a computer are as follows:

- (a) Input Unit
- (b) Central Processing Unit (CPU)
- (c) Output Unit
- (d) Memory Unit

4. Who invented the mouse?

Ans. Mouse is a pointing (input) device that was developed by Douglas Engelbart in 1963.

5. What are the two parts of main memory?

Ans. Two parts of main memory are as follows:

- Random Access Memory (RAM)
- Read-Only Memory (ROM)

6. Why are the secondary storage devices required?

Ans. The secondary storage devices are required because of the following two reasons:

- The primary memory is finite and limited in size.
- In secondary storage, data and programs do not disappear when power is turned off.

7. Two devices used by the supermarket point-of-sale (POS) terminal are a bar code reader and a keyboard. Name two other input/output devices used at the POS and give a use for each device.

Ans. (a) Monitor—used to display the information about the items purchased or sold.
(b) Printer—used for taking printout of the bill or invoice generated.

8. Name the input or output device used to do the following:

- (a) To output audio
- (b) To build 3D models
- (c) To enter audio-based commands

Ans. (a) Speaker/Earphones/headphones- (b) 3D Printer
- (c) Microphone (Mic)

9. Given below are some features of two types of computer memories—RAM and ROM. List each feature under RAM or ROM.

- (a) Non-volatile memory
- (b) Contents can't be changed
- (c) Stores data or files the user is currently working on
- (d) Volatile memory
- (e) Can be written to and read from

Ans. RAM: (c), (d), (e);

ROM: (a), (b)

10. What does a bus mean?

Ans. A bus is a group of conducting lines that carries data, address and control signals between CPU and memory.

11. Why is data bus bi-directional?

Ans. Data bus is bi-directional because same bus is used for data transmission from microprocessor to memory location or input/output device and vice versa.

12. Name the part of computer which is responsible for processing.

[HOTS]

Ans. The CPU is responsible for processing.

13. Name the primary memory which is volatile.

Ans. RAM (Random Access Memory).

14. Write down the various capabilities of computer.

Ans. The various capabilities of computer are: speed, reliability, diligence, versatility and large memory.

15. What is booting?

Ans. Booting is the process of starting your computer and loading the operating system.

16. What do you understand by the terms hardware and software?

Ans. The components that can be physically seen and touched are called hardware whereas software is a set of computer programs and related data that provide a list of instructions to govern computer operations in an organized way.

17. Which of the following is the function(s) of the operating system?

- (a) Processor
- (b) Memory Management
- (c) File Management
- (d) All of these

Ans. (d) All of these.

18. Name the two categories of system software.

Ans. System software are divided into the following two categories:

- (a) Operating system
- (b) Language processor

19. Why are language processors used? List their types.

Ans. Programs written in a high-level language are required to be converted into a machine language. This task is accomplished by the language processor. The types of language processors are as follows:

- (i) Assembler
- (ii) Interpreter
- (iii) Compiler

20. Explain briefly the functions of Operating System.

Ans. The main functions of OS are:

- (i) Memory management
- (ii) Processor management
- (iii) Device management
- (iv) Information management

21. What types of operations are provided by an operating system?

Ans. The operations provided by an operating system are:

- Provide an interface to allow the user to communicate with the computer.
- Control the various system hardware and software resources, e.g., Keyboard, Visual Display Unit (VDU), Printer, etc., using special programs called Device Drivers.
- Manage the user's files, keeping track of their positions on the disk, updating them after the user makes changes, etc.
- Provide system facilities, e.g., Compilers to translate from high-level programming languages used by the user to the internal machine language the computer uses.

22. What is the function of memory? What are its measuring units?

Ans. The memory temporarily holds data and information during processing. The smallest unit of memory is a byte (8 bits). A byte can store one character in binary form. Other measuring units are kilobyte (KB) equal to 1024 bytes, Megabyte (MB) equal to 1024 KB, Gigabyte (GB) equal to 1024 MB and Terabyte (TB) equal to 1024 GB.

23. What is the difference between RAM and ROM?

Ans. RAM refers to random access memory where both read and write operations can take place. But RAM is a volatile memory; its contents are lost when power is turned off. ROM refers to read-only memory where only read operation can take place. ROM is a non-volatile memory. Both RAM and ROM are parts of the primary memory.

24. Give examples of application software and system software.

Ans. Application software such as Word processors (MS Word, OpenOffice.org-Writer), spreadsheet program (OpenOffice.org Calc, MS Excel), database program (MS Access, OpenOffice.org Base) perform productive tasks for users while system software such as operating systems (Windows XP, Vista), interface with hardware systems.

25. What is the difference between a compiler and an interpreter?

Ans. The difference between a compiler and an interpreter is described as follows:

Interpreter translates a program written in a high-level language into machine language by converting and executing it line by line. The interpreter is very useful for error-debugging as it displays errors while translating a program into machine language. It cannot execute a program until all the errors are resolved.

Compiler works the same way as an interpreter. However, the main difference between an interpreter and a compiler is that compiler converts the entire program into machine language in one go and also reports all the errors in the program along with the line numbers. When all the errors are rectified, the program is recompiled and after that the compiler is no longer needed in the memory.

26. Explain operating system and utility system software. Give examples of utility programs.

Ans. Operating system is a software program which manages the hardware and software resources and links them. It creates an interface between hardware and software. It is a bridge between hardware and software.

Operating systems are essential for running a computer. However, as part of the operating system, there are some programs which are not essential for the running of the computer. These are known as utility programs which perform specific tasks related to the management of computer functions. If they were not there, the working of the computer would not be affected at all.

Some examples of utility programs are:

- (i) Virus scanners
- (ii) Disk defragmenters
- (iii) Encryption utilities
- (iv) Backup software

27. Why is primary memory termed as 'destructive write' memory but 'non-destructive read' memory? [HOTS]

Ans. When a memory location is read from the primary memory, the contents of the memory word remain the same; they are not altered. Therefore, a primary memory is termed as 'non-destructive read' memory since the read operation does not destruct the contents of a memory word.

When a write operation takes place, the previous contents of the memory word are overwritten. Thus, the primary memory is termed as 'destructive write' memory as the write operation destructs the contents of a memory word.

28. What is the difference between an address bus and a data bus?

Ans. The difference between an address bus and a data bus is as follows:

Address Bus: An address bus contains the address of the piece of memory or I/O devices to be read from or written to. One wire is required for each bit, which means 16 bits will require 16 wires. A 16-bit binary number allows 2^{16} or 32,000 different numbers.

Data Bus: A data bus actually carries the data to be processed in the binary form. It carries the information between the processor and various other external units, such as memory. Its typical size is 8 or 16 bits.

29. Explain the usage of the smart card reader.

Ans. We are aware of ATM cards that are used in ATM machines. ATM cards store data through magnetic strips, which are pasted on the back of these cards. Magnetic strips hold much more data than printed characters or bar codes per unit of space. These strips are not human-readable and are, therefore, used to store confidential data. Moreover, it is almost impossible to create a duplicate copy of the cards containing such strips. Special reader machines are required to decode the encoded data stored on these magnetic strips.

The enhanced version of a card containing magnetic strips is called smart card. A smart card is embedded with a microprocessor that can hold a certain amount of personal data in its memory. The special reader machine that is used to decode the data on smart cards is known as smart card reader.

30. What is the difference between data and information?

Ans. The difference between data and information is as follows:

Data is defined as raw facts and figures such as "Tejas", "MBA", "2013", "ABC". This data does not have any meaning when presented as such. However, this data can be organized or processed to transform it into useful information.

Information is defined as a collection of data which is organized in a particular manner to generate some meaning. For example, "Tejas passed MBA in 2013 from ABC" is the information that we got after processing the data given above.

31. Define the following terms:

- (i) Application Software
- (ii) Cache Memory
- (iii) Language Processor

Ans. (i) **Application Software:** This software is needed to cater to different requirements of the end-users. This specific software that works on top of the system software is termed as application software, for example, MS Word.

(ii) **Cache Memory:** To speed up the operations of the CPU, a very high speed memory is placed between the CPU and the primary memory, known as cache. It stores copies of the data from frequently accessed primary memory locations, thus reducing the average time required to access data from the primary memory.

(iii) **Language Processor:** The software required to convert a program written in High Level Language to machine code is known as Language Processor. It is of three types—Compiler, Interpreter and Assembler.

32. Identify the category (system or application) of the following software:

- Ans.
- (a) Interpreter – System Software (Language Processor)
 - (b) Backup Software – System Software (Utility Software)
 - (c) PowerPoint – Application Software
 - (d) Linux – System Software (OS)
 - (e) Compiler – System Software (Language Processor)
 - (f) Antivirus – System Software (Utility Software)
 - (g) WinRAR – System Software (Disk Compression Utility Software)

UNSOLVED QUESTIONS

1. Define a computer.
2. How does an ALU work?
3. Briefly explain the working of a control unit.
4. Define hardware and software.
5. What is an operating system?
6. What is software?
7. List some of the hardware in computer equipment.
8. Explain the types of operating systems with examples.
9. "Hardware is of no use without software and software cannot be used without hardware." Explain.
10. How can software be classified? Name at least one software in each of the categories.
11. What is an operating system? Write names of any two popular operating systems.
12. Specify the measuring units of memory.
13. Identify the category (system, application, programming tool) of the following software:
 - (a) Compiler
 - (b) Assembler
 - (c) Ubuntu
 - (d) Text Editor

14. What do you mean by memory devices? Explain RAM and ROM.
15. What are output devices? Give some examples.
16. List different types of impact printers.
17. Briefly explain the working of a laser printer.
18. What is the use of a light pen?
19. What do you understand by RAM and ROM?
20. Given below are some features of computer memories—RAM and ROM. List the below-given features under RAM or ROM:
 - (a) Non-volatile Memory
 - (b) Content can't be changed
 - (c) Volatile Memory
 - (d) Can be written to and read from
21. Expand the following term:
 - (a) ALU
 - (b) GUI
22. Explain, in brief, the working of a touchscreen.
23. What are plotters? List different types of plotters.
24. Check the size of RAM and HDD of a computer in your school. Make a table and write their size in Bytes, Kilobytes, Megabytes and Gigabytes.
25. List all secondary storage devices available at your school or home.
26. Which operating system is installed on your computer at home or in school?
27. Convert the following into bytes:
 - (a) 2 MB
 - (b) 3.7 GB
 - (c) 1.2 TB
28. Identify the type of software:
 - (a) The software acts as an interface between a user and the hardware.
 - (b) It converts the program written in HLL into machine language by converting and executing it line by line.
29. Write down the type of memory needed to do the following:
 - (a) To store data permanently
 - (b) To execute the program
 - (c) To store the instructions which cannot be overwritten.
30. How is an interpreter different from compiler on these parameters?
 - (a) Memory requirement during translation
 - (b) Memory requirement after translation
 - (c) Number of instructions translated
31. What is the major difference between optical and magnetic discs?
32. What do you mean by a communication bus? List its types.
33. Define port along with some examples.
34. List the differences between a CD and a DVD.
35. Explain the statement, "Functioning of a computer is similar to the way the human brain functions."
36. What is the role of Graphics Processing Unit (GPU) with regard to smartphone communications processor?
37. Explain, in brief, the basic architecture of a computer.
38. What is an input unit? Give its significance.

39. List and briefly explain all the components of a CPU.
40. Why is a control unit referred to as the central controller of a computer?
41. Does an ALU work independently or in coordination with some other unit? Give reasons.
42. Compare data and information.
43. How does an output unit work? Give examples of some output devices.
44. What is the role of memory in the functioning of a computer?
45. List all the hardware components of a computer. Give their significance.
46. Explain various components of a computer system and show the relationship between them with the help of a diagram.
47. Compare volatile memory and non-volatile memory.
48. Define primary storage devices. List their types.
49. What do you mean by a mouse? List its types.
50. What is a bar code? Why is a bar code reader used?
51. Explain, in brief, inkjet and laser jet printers.
52. What is a plotter? Give the benefits and limitations of using plotters.
53. Explain any three types of input devices.
54. Explain any three types of output devices with their purpose.
55. Define VDU. Name its two types.
56. Give the difference between an impact and a non-impact printer.
57. Discuss the following devices:
 - (a) Light Pen
 - (b) Graphic tablet
 - (c) CD-ROM
 - (d) DVD
58. What is the significance of address and data buses?
59. Describe the basic architecture and functioning of a computer.
60. Describe the terms hardware and software along with their components.
61. Explain plotters and their types.
62. Discuss the various categories of printers.
63. What do you mean by memory devices? Explain RAM and ROM.
64. Discuss the classification of digital computers.
65. Discuss secondary storage devices in detail.
66. What are memory cards? Discuss their types.
67. Explain the following:
 - (a) Assembler
 - (b) Compiler
 - (c) Interpreter

CASE-BASED/SOURCE-BASED INTEGRATED QUESTIONS

1. Xenox Technologies deals in hardware components required for assembling computer systems in the Nehru Place market. They provide reliable and efficient data storage devices to their customers. Six storage devices in which they deal are described below. Name the storage device being described and also list the appropriate category of storage.
 - (a) Optical media which use one spiral track; red lasers are used to read and write data on the media surface; makes use of dual-layering technology to increase storage capacity.
 - (b) Non-volatile memory chip: contents of the chip cannot be altered; It is often used to store the start-up routines in a computer (e.g., the BIOS).

- (c) Optical media which use concentric tracks to store data; this allows read and write operations to be carried out at the same time
- (d) Non-volatile memory device which uses flash memories (which consist of millions of transistors wired in a series on a single circuit board).
- (e) Optical media which use blue laser technology to read and write data on the media surface.
- (f) Magnetic disc with very large storage capacity; can be used to store vast amounts of data; mostly fixed in computer cases and serves as the main storage device.

Ans. (a) DVD	Offline Storage
(b) ROM	Primary Memory
(c) DVD-ROM	Offline Storage
(d) Solid State Memory/Memory Card	Offline Storage
(e) Blu-ray Disc	Offline Storage
(f) Hard Disk	Secondary Memory.

2. World Class Telecom is planning to have a videoconference with its associates across several nations. In its videoconferencing area, there are some microphones, speakers and webcams. Describe the purpose of each of the devices.

Ans. **Microphone:** This device is used to input sound or the vocal input of the people speaking.

Speakers: These are needed so as to hear the sound/audio output from the computer.

Webcam: Webcam or web camera is needed to capture the video image of the scene and all the people participating in videoconferencing.

3. ABC School newsletter contains text and images in it. The head teacher needs one thousand copies of this newsletter. Give four reasons why a laser printer should be used rather than an inkjet or a dot matrix printer.

- Ans.
- High quality printouts—better than inkjet or dot matrix.
 - Fast printouts—faster than inkjet or dot matrix.
 - Makes very little noise—much less than inkjet or dot matrix.
 - Low per page cost—cheaper than inkjet or dot matrix.